DPUAmplifier

1. Inputs:

In	Input value
Multiplicator	Gain of the amplifier. Default: 1
Divisor	Attenuation of the amplifier. Default: 1
Offset	Offset correction at the amplifier output. Default: 0

2. Dynamic parameters:

OutMin	Lower bound of the output. Default: -1^{300} (no limit)
OutMax	Upper bound of the output. Default: 1^{300} (no limit)
Stiffness	Exponent for non linear output characteristics. Default: 1
StiffnessGain	Gain for non linear output characteristics. Default 1
StiffnessOffset	Offset correction for non linear output characteristics. Default: 0

3. Outputs:

Out	Output of the amplifier.

4. Functional principle:

The output of the amplifier is calculated by the following equation:

$$Out = \frac{In \cdot Multiplicator}{Divisor} + Offset$$

Using the stiffness parameters nonlinear characteristics can be achieved. This is based on the following equation:

$$Out = \left(|Out|^{Stiffness} \cdot sgn(Out)\right) \cdot StiffnessGain + StiffnessOffset$$

The output value is limited to the range $[OutMin, OutMax]$.

5. Example:

Convert the speed from meters per second to kilometers per hour:

```
DPUAmplifier Convert
{
	Computer = {VDYN};
	Index = 1;

	Multiplicator = 3600; # seconds to hour (60 * 60)
	Divisor = 1000 # meters to kilometers
```

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```
	# can be easily done with only
	# Multiplicator = 3.6;
};

Connections = {
	VDyn.v -> Convert.In
};
```

Invert a 0/1 signal:

```
DPUAmplifier Invert
{
	Computer = {LOCALHOST};
	Index = 1;

	Multiplicator = -1;
	Offset = 1
	OutMin = 0;
	OutMax = 1;
};
```